

Boundary and initial conditions in EXHALE (EXHALE)

This memo documents the boundary conditions and the initial conditions of the EXHALE / EXHALE hydrodynamics solver at the level of the actual implementation: what is imposed, where in the code, why it has the form it does, and what dynamical behavior it produces (in particular the “breathing base”). File references are relative to the repository root; the structure is inherited from ATES v2 (Caldirola et al.) unless noted as an EXHALE extension.

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1. Overview: where and when the BCs act

1.1 Ghost-cell layout

The radial grid carries N computational cells ($j = 1, \dots, N$) plus $N_g = 2$ ghost cells on each side, so all state arrays are dimensioned $(1-N_g : N+Ng)$, i.e. $j = -1, 0$ below the base and $j = N+1, N+2$ above the top (`src/modules/init/parameters.f90`, `define_grid.f90`).

2026-08-15: the cell count is a runtime quantity. N used to be a compile-time parameter fixed at 500. It is now the module variable `integer :: N = 500` of `parameters.f90`, set by the optional `input.inp` key `Grid cells: <N>` (`input_read.f90`); an `input.inp` without the key keeps 500 and reproduces its earlier result. The grid-sized module arrays became allocatable (lower bound $1-N_g$ unchanged) and are allocated by `allocate_grid_arrays` once parsing has resolved the key, so nothing below is affected other than that N may now differ from 500. Values under 10 are refused. Two consequences appear later: the base refinement pairs of §2.5, and the restart record count of §5.5.

The first interior cell $j = 1$ sits at the lower boundary radius $r = 1$ (lengths are normalized to the planetary radius R_0); the last interior cell $j = N$ sits at $r_{\max}$, which is either the Roche/Hill truncation $r_{\max} = (3M_*/M_p)^{-1/3} \tilde{a}$ in Roche mode or a user-chosen outer radius in spherical mode (`input_read.f90`).

All boundary conditions are applied to the *ghost cells only*; the interior cells $1, \dots, N$ are never overwritten. The fluxes at the two boundary interfaces then emerge from the same HLLC Riemann solver used everywhere else, with the ghost states as the outside input. In this sense every BC in the code is a *weak* (flux-mediated) condition: even the “Dirichlet” base values act on the interior only through the Riemann problem at the interface $j = 1/2$.

1.2 The three entry points (`Apply_BC.f90`)

All BC logic lives in one module, `src/modules/states/Apply_BC.f90`:

- `Apply_BC(u,u)` — conservative-variable wrapper: converts $U \rightarrow W$ (primitive), calls `Apply_BC_W`, converts back. Called after every Runge–Kutta stage in the main loop (`EXHALE_main.f90`), before every residual evaluation, and by the steady-state Newton solver each time it unpacks a trial state.
- `Apply_BC_W(W,W)` — the actual assignment of the four ghost states: lower ghosts via `BC_component_constrho` (§2.), upper ghosts via extrapolation (§3.).
- `Rec_BC(WL,WR,...)` — the same conditions applied to the *reconstructed* left/right interface states after PLM or WENO3 reconstruction (`Reconstruction.f90`), so that the boundary stencils of the higher-order scheme see BC-consistent data rather than raw extrapolated slopes.

In the Newton steady-state solver the ghost cells are deliberately *not* unknowns: the residual is defined on the interior state only, and `Apply_BC` regenerates the ghosts from the interior at every function evaluation

(src/modules/time_step/steady_newton.f90). The BCs are therefore identical between the time-marching and Newton phases by construction.

1.3 Normalization constants tied to the base

The lower BC is intertwined with the code-wide normalization (input_read.f90):

$$\mu = \frac{1+4Y}{1+Y} m_H, \quad v_0 = \sqrt{\frac{k_B T_0}{m_H}}, \quad p_0 = n_0 m_H v_0^2 = n_0 k_B T_0, \quad b_0 = \frac{GM_p m_H}{k_B T_0 R_0}, \quad (1)$$

where $Y \equiv n_{\text{He}}/n_{\text{H}}$ is the He/H number ratio from input.inp, n_0 is the user-specified total *nuclei* number density at the base, and T_0 the base temperature. (The code variable mu is m_H itself; the composition factor appears separately as rho_bc, see below.) Densities are stored as mass densities in units of $n_0 m_H$, number densities in units of n_0 , pressures in units of p_0 , temperatures in units of T_0 , velocities in units of the isothermal hydrogen sound speed v_0 .

The equation-of-state closure used to convert between the hydrodynamic pressure and the temperature handed to the ionization/radiation module is the dimensionless ideal-gas law including free electrons (composition.f90):

$$T = \frac{P}{n_{\text{tot}} + n_e}, \quad (2)$$

with n_{tot} the total *atomic* (nuclei) number density and n_e from calc_ne. Under the default eos_metals 1 policy (2026-06-13) both include the trace metals — their nuclei in n_{tot} and their electrons in n_e — consistently with the charge balance inside the ionization solver; eos_metals 0 in metals.inp restores the legacy H/He-only budget (§2.7).

2. Lower boundary: fixed reservoir with a one-way valve

The two lower ghost cells are set by BC_component_constrho (Apply_BC.f90, lines 31–80). In the cell-index-first notation of this memo (W(cell, component); the code stores the transpose):

```
W(ghost,1) = rho_bc ! density: Dirichlet
if (base_v_massflux .and. base_flux_const > 0) then ! mass-flux base
  W(ghost,2) = base_flux_const/(rho_bc*r(ghost)**2)
else if (valve_eps > 0) then ! smooth valve
  W(ghost,2) = 0.5*( W(1,2) + sqrt(W(1,2)**2 + valve_eps**2) )
else ! legacy hard valve
  W(ghost,2) = max( W(1,2), 0.0 )
endif
if (hydrostatic_base) then ! dp/dr continuous
  W(ghost,3) = W(1,3) + (W(2,3) - W(1,3))/(r(2) - r(1))*(r(ghost) - r(1))
else if (base_ghost_T_continuous) then ! dT/dr = 0
  W(ghost,3) = (ntot_bc + dp_bc)*W(1,3)/n_part_cell1
else ! T = T0 (default)
  W(ghost,3) = ntot_bc + dp_bc
endif
```

The default closures (legacy valve, isothermal ghost pressure) give both ghost cells the *same* state, so there is no gradient across them; the mass-flux velocity and the hydrostatic ghost pressure are evaluated at $r(\text{ghost})$ and therefore do differ between $j = 0$ and $j = -1$. The three components implement three distinct ideas, discussed in turn; the opt-in velocity and pressure branches are collected in §2.6.

2.1 Density: a fixed-composition reservoir

The dimensionless ghost density is the constant

$$\rho_{\text{bc}} = \frac{m_{\text{gas}}/\text{H}}{1+Y}, \quad m_{\text{gas}}/\text{H} = 1+4Y + \sum_X a_X A_X, \quad a_X \equiv \frac{n_X}{n_{\text{H}}}, \quad (3)$$

the mean gas mass per H/He nucleus in units of m_{H} (`comp_rho_bc()` = `comp_mass_per_H()`/(1+ Y) in the code). Because mass densities are normalized by $n_0 m_{\text{H}}$, this pins the ghost cells to “H/He nuclei density = n_0 with the prescribed composition”: the hydrogen nuclei density at the base is $n_{\text{H}} = n_0/(1+Y)$, the helium density Y times that, and each metal $a_X n_{\text{H}}$. The base therefore behaves as an *infinite isochoric reservoir*: however much mass the wind removes, the boundary cell always refills to the same density.

The metal mass term $\sum_X a_X A_X \approx 0.016$ at solar abundances (a $\sim 1.2\%$ effect) is included by the default `eos_metals 1` policy, which also adds the metal mass to `calc_rho`; `eos_metals 0` restores the legacy H/He-only trace approximation, $\rho_{\text{bc}} = (1+4Y)/(1+Y)$. That legacy expression is therefore *only* the value of a metals-off atomic base; it is not the general form.

One source for the base composition. ρ_{bc} , m_{gas}/H (`mass_per_H`) and $n_{\text{tot,bc}}$ (`ntot_bc`) are all produced by the three functions `comp_rho_bc`, `comp_mass_per_H` and `comp_ntot_bc` of `src/modules/functions/composition.f90`, which `input_read` calls once; that module is the single place where the base composition policy lives, so no code path can disagree with another about what the base is made of. Three switches move the numbers inside it:

- `eos_metals` (`metals.inp`; default 1 since 2026-06-13) puts the trace-metal mass into m_{gas}/H and the trace-metal nuclei into $n_{\text{tot,bc}} = (1+Y + \sum_X a_X)/(1+Y)$. With `eos_metals 0`, or with no metals, both collapse to the H/He values $1+4Y$ and 1.
- `Molecular base`: `True` (Tier-2a; auto-enabled whenever the molecular chemistry is on) subtracts from $n_{\text{tot,bc}}$ the particles lost to the H nuclei bound into H_2 at the base: two H nuclei make one molecule, so a bound H fraction x_2 costs $x_2/2$ particles (`h2_bound_fraction`). This changes the particle count only — ρ_{bc} , hence the mass reservoir, is untouched, and the species arrays stay atomic.
- A `base.inp` handoff supplies q_{H_2} at the base (`q_H2_base`) and then takes precedence over the chemical-equilibrium fit that `h2_mixing_ratio_base` otherwise evaluates at (p_{base}, T_0) ; `input_read` echoes which of the two set $n_{\text{tot,bc}}$. Rationale: `docs/base_composition_handoff_plan.md`.
- The same handoff can move the *reservoirs* these formulas are built from, not only the particle count: `HeH_base` sets Y , and one `<El>_H_base` key per element (any of the ten symbols of `species_table`, e.g. `O_H_base`) sets the corresponding a_X , overriding `metals.inp` for that element and turning the metal system on by itself when

it is the only nonzero abundance. `thereis_metals`, `melem_ab` and `thereis_lowIP_metal` are all derived *after* the handoff, so a handoff element enters ρ_{bc} , m_{gas}/H and $n_{\text{tot},bc}$ exactly as a `metals.inp` element does.

- A lower-atmosphere *profile* (`Lower_atmosphere_profile`:, `lower_atmosphere_profile.f90`) states T_0 , R_0 , p_{base} , q_{H_2} , Y and every elemental ratio it carries at its own matching pressure, and is read before `read_base_inp` and applied after it, so nothing scalar can overwrite a profile value. With one in use the EOS-boundary, elemental-reservoir and boundary-constraint keys of `base.inp` are refused by name. The columns `n_tot` and `rho` of the profile are carried but *not* imposed: the base density stays the n_0 of `input.inp` and $n_{\text{tot},bc}$ still follows from T_0 , Y , q_{H_2} and the elemental ratios through `comp_ntot_bc`, exactly as on the scalar path. Schema: `docs/input_schema.md` §2d.

2.2 Pressure: dynamically updated to pin the base temperature

The ghost pressure is $p_{\text{gh}} = n_{\text{tot},bc} + \delta p_{bc}$, where $n_{\text{tot},bc} = (1 + Y + \sum_X a_X)/(1 + Y)$ is the (constant) total nuclei density at the base in units of n_0 (`ntot_bc`; = 1 for H/He only, and reduced by the H_2 -bound share under a molecular base) and δp_{bc} starts at the placeholder 10^{-10} and is recomputed after every ionization-equilibrium solve as the base electron density (`ionization_equilibrium.f90`):

$$\delta p_{bc} = \frac{n_{\text{HII}} + n_{\text{HeII}} + 2n_{\text{HeIII}} + \sum_i q_i n_i^{\text{met}}}{n_0} \Big|_{j=1-N_g} = \frac{n_e(\text{base})}{n_0}, \quad (4)$$

with q_i the metal ion charges (included under `eos_metals 1`, dropped under `eos_metals 0`). The point of this correction is visible through the EOS closure (2): at the ghost cells $n_{\text{tot}} = n_{\text{tot},bc}$ (by the density BC) and $n_e = \delta p_{bc}$, so

$$T_{\text{gh}} = \frac{n_{\text{tot},bc} + \delta p_{bc}}{n_{\text{tot},bc} + \delta p_{bc}} = 1 \quad \Longleftrightarrow \quad T = T_0. \quad (5)$$

In other words, the pair (fixed ρ , dynamically corrected p) is really the pair (*fixed nuclei density* n_0 , *fixed temperature* T_0): as photoionization develops at the base and free electrons add partial pressure, the ghost pressure is raised by exactly the electron share so that the base stays isothermal at T_0 instead of being artificially cooled. For a near-neutral base (the usual situation, since the base optical depth is huge) δp_{bc} is tiny and the BC reduces to $p = p_0$.

The metal-electron share of δp_{bc} is small (even with all low-ionization-potential metals singly ionized at the base it is $\lesssim 4 \times 10^{-4}$), but it is carried anyway under `eos_metals 1` for consistency with `calc_ne`.

2.3 Velocity: the one-way valve

The ghost velocity copies the first interior cell when the gas moves outward and is clamped at zero when it moves inward:

$$v_{\text{gh}} = \max(v_1, 0). \quad (6)$$

Neither a wall ($v_{\text{gh}} = 0$) nor a free condition ($v_{\text{gh}} = v_1$) would be physical here:

- a *wall* would starve the wind — the escaping flow is launched by photoionization heating *above* the base, and the mass it carries must be resupplied through the lower boundary;

- a *free* condition combined with the fixed-density ghosts would let the reservoir act as an unphysical mass sink whenever the interior gas sags downward, swallowing arbitrary amounts of gas at fixed ρ_{bc} .

The valve resolves the asymmetry: *outflow is allowed (the reservoir feeds the wind), sustained inflow is refused (the reservoir does not absorb a backflow)*. Physically the boundary models the top of an essentially infinite, hydrostatic lower atmosphere held at (n_0, T_0) .

Smooth valve (EXHALE extension). The hard max has a kink at $v_1 = 0$ — and, as discussed in §2.4, $v_1 = 0$ is precisely where the base dynamics lives. The kink is invisible to the explicit time-marching scheme but fatal to the steady-state Newton solver, whose line search cannot cross a non-differentiable point of the residual. The input key `Valve eps`: therefore enables the smooth approximation

$$v_{gh} = \frac{1}{2} \left(v_1 + \sqrt{v_1^2 + \varepsilon^2} \right), \quad (7)$$

which is C^∞ , tends to v_1 for $v_1 \gg \varepsilon$, tends to 0 for $v_1 \ll -\varepsilon$, and biases the valve by at most $+\varepsilon/2$ near $v_1 \sim 0$. With $\varepsilon = 0$ (default) the legacy hard valve is used.

2.4 The “breathing base”

Because the base velocity is a free variable governed by an *asymmetric* boundary (an open pipe outward, a wall inward), the near-base flow does not settle onto the exact $v = 0$ fixed point but onto a small limit cycle around it. One cycle proceeds as follows:

1. **Exhale** ($v_1 > 0$): the wind draws mass up; the Riemann problem against the fixed- ρ_{bc} ghosts supplies gas at base density — the boundary acts as a mass source.
2. **Overshoot**: if momentarily more mass is lifted than the heating profile above can carry, the over-supplied column begins to sink back under gravity; v_1 turns negative.
3. **Inhale** ($v_1 < 0$): the valve closes ($v_{gh} = 0$). The descending gas piles up on the closed boundary, raising ρ and p in the first few interior cells — a transient wall reflection.
4. **Rebound**: the accumulated pressure pushes the column back out; v_1 turns positive and the cycle repeats.

This inhale/exhale oscillation is the *breathing base*. It is an intrinsic consequence of the valve-type BC — present in the original ATES as well — and not a numerical defect. Two regimes should be distinguished (see docs/initial_condition_benchmark.pdf, §7):

Self-limiting (normal). In the time-marching hydrodynamics the pressure built up during the inhale phase immediately produces the restoring force of step 4, so the amplitude saturates. The oscillation stays confined to the first few cells ($r \lesssim 1.1 R_p$) and has no measurable effect on the wind region or on $\dot{M}$.

Self-amplifying (pathological). If the *global* initial condition is inconsistent (the canonical example: a naive full-Parker density profile, whose hot column drains onto the base), each cycle deposits more mass than the previous one and the breathing grows until the run diverges. The cure is a consistent IC (hydrostatic density, optionally with a warm seed overlay), not a change of the BC.

Numerical consequences. Harmless as it is physically, the breathing base interacts with three parts of the code, each of which carries a dedicated counter-measure:

- *Convergence monitoring.* The breathing cells never stop oscillating, so a convergence criterion of the form $\max_j |\Delta u|$ over the whole grid can stall at the 10^{-2} – 10^{-1} level indefinitely. ATES therefore evaluates its momentum-flatness and du/dtu diagnostics only over the wind region $[j_{\min} : N]$, where $j_{\min}$ is the first cell with $r \geq r_{\text{esc}}$ (the Escape radius input; §4.2). Cases with unusually violent breathing (HD 189733 b is the standing example) can still fail the strict du threshold; reducing CFL, modifying the valve, or replacing the boundary Riemann solver did *not* remove the oscillation in our experiments — it is robustly attached to the BC type. A periodic Shapiro (1970) low-pass filter, the stabilizer the reference code CETIMB (Koskinen et al. 2013a) applies to exactly this gravity-unbalanced sound-wave instability, *does* damp the oscillatory component (§2.6), but it biases the momentum and pushes the clean cold-IC cases off the wind branch into infall, so it is kept opt-in rather than a default.
- *Newton solver.* The $v_1 = 0$ kink of the hard valve sits exactly on the breathing orbit; the smooth valve of §2.3 exists for this reason.
- *Advection post-processing.* The `_adv` correction integrates ionization states along streamlines; during the inhale phase ($v \leq 0$) the streamline of a base cell exits the domain through the lower boundary and the correction is ill-defined (this produced the historical temperature spike/sawtooth at $r \approx 1.1$). The post-processor therefore keeps the equilibrium solution and skips the advection correction wherever $v \leq 0$ at the base.

2.5 Base grid resolution (2026-08-11)

Not every alternating pattern at the base is the breathing limit cycle. On the Mixed grid the base region is N_{low} uniform cells of width Δr_{base} stacked on $r = 1$, and the physical requirement on that width is that it resolve the base density scale height $H = kT/(\mu g)$. Where $H/\Delta r_{\text{base}}$ is of order a few cells, the discretization supports a *stationary* $2\Delta r$ entropy (contact) mode that nothing in the scheme damps — HLLC resolves a contact of zero speed exactly, the gravity source is cell-local, and the WENO3 pressure gradient sees only interface pressures. The measured dependence is an alternating amplitude in $\ln \rho$ of $\sim 10^{-2}$ for $H/\Delta r < 5$ against $\sim 10^{-4}$ for $H/\Delta r > 100$ (docs/hd189_base_checkerboard.md). High-gravity planets have the least margin: on the default grid `write_setup_report` echoes 26.9 cells per H at T_{eq} for HD 189733 b against 102.3 for WASP-121 b.

The resolution is therefore an input key, `Base grid [dr,cells]: <dr_base> [<N_low_cells>]` (`input_read.f90`; defaults `2.0e-4 50`, ignored by the Uniform and Stretched grid types). The two numbers are not independent — their product is the radial extent of the uniform region, $0.01 R_p$ with the defaults — so refining the base at fixed extent means dividing Δr_{base} and multiplying N_{low} by the same factor. Because those extra base cells are taken out of the same total N , the runtime `Grid cells: key of §1.` is what lets a refinement keep the outer stretched region unchanged; the paired settings are

Base grid [dr,cells]:	Grid cells:
1.0e-4 100	500
5.0e-5 200	658
2.5e-5 400	916
1.25e-5 800	1375

Note that spelling the default out explicitly (`Base grid [dr,cells]: 2.0e-4 50`) does *not* reproduce a run made without the key: the compiled-in default is a default-real literal, 2.5×10^{-8} relative below the exact double the list-directed read produces, which moves every base cell by that amount. The full measurement — what sets the mode, which remedies were tried, and what each grid step bought — is `docs/hd189_base_checkerboard.md`.

2.6 Optional stabilizers and the selectable base mode (EXHALE)

EXHALE adds six opt-in controls of the base behavior. The first three were ported while studying the breathing/infall behavior against CETIMB (Koskinen et al. 2013a); the Shapiro filter in particular was adopted after Huang et al. (2023) — whose escape model is built on CETIMB — pointed to CETIMB’s use of exactly this low-pass to damp the gravity-unbalanced base sound waves; that reference is what suggested trying it here. The last three (2026-08) came out of the HD 189733 b base checkerboard and the HD 209458 b stagnant-layer investigations. All six *default to the legacy behavior* — the numeric ones are off at $\varepsilon \leq 0$, the ones selected by name default to valve, density and isothermal — so a run that does not ask for them is unchanged. The measured effects of the first three are tabulated in §7..

Two of them — `Hydrostatic base` and `Base ghost temperature: continuous` — set the *same* quantity, the ghost pressure, and are therefore mutually exclusive: `Hydrostatic base` is tested first in `BC_component_constrho` and `input_read` warns when both are given.

Shapiro filter (`Shapiro filter: <eps> [<every>]`). Every every steps a weak two-step Shapiro (1970) 1–2–1 low-pass filter $u_j \leftarrow \frac{\varepsilon}{4}(u_{j-1} - 2u_j + u_{j+1})$ is applied to the conservative variables. This is CETIMB’s cure for the gravity-unbalanced sound waves and it does smooth the breathing transient. *However*, on the otherwise-clean cold-IC cases it injects a systematic momentum bias that drives the solution off the transonic-wind saddle into the infall branch ($v < 0$ almost everywhere; see the `_Shap` rows of Table 3). It is therefore opt-in — useful only for cases that genuinely breathe, not as a global default.

Mass-flux base velocity (`Base velocity: massflux`). Instead of the one-way valve (§2.3), the base velocity is set from the steady mass-flux continuity $v_0 = F_c / (\rho_0 r_0^2)$ with F_c a slow exponential moving average of $\rho v r^2$ over the wind region. It is physically benign — it never causes infall on its own and gives the cleanest base (it suppresses the $\pm m s^{-1}$ valve oscillation) — but it slows convergence ($\sim 5 \times$ more steps), so the fast valve stays the default.

Pressure-anchored base (`Base BC: pressure [<p_ubar>]`). Instead of fixing the base *number density* at n_0 (§2.), this fixes the base *pressure* at `p_ubar` microbar (default $1 \mu\text{bar}$, the CETIMB lower boundary) and derives n_0 from $p = n_0 k_B T_0 n_{\text{tot},bc}$. Because the isothermal base is over-determined (it already pins $T = T_0$), this is a reparameterization of the same ghost; its physical effect is simply a much less dense base ($1 \mu\text{bar} \Rightarrow n_0 \approx 5 \times 10^{12}$, $\sim 20 \times$ below 10^{14}), which shrinks the dense-base gravity source ρg that feeds the breathing. Empirically a

density-anchored run at the same low n_0 ($\log n_0 = 12.5$) gives an almost identical converged wind, confirming that the lever is the base *density*, not the anchoring variable (Tables 2 and 3).

Base ghost temperature (Base ghost temperature: isothermal|continuous, 2026-08-11). Selects the temperature closure of the lower ghost cells; the density anchor ρ_{bc} is untouched either way. *isothermal* (default, legacy) is the pin of §2.2: the ghost pressure is held at $n_{\text{tot},bc} + \delta p_{bc}$, i.e. $T_{\text{gh}} = T_0$. *continuous* imposes $dT/dr = 0$ at the base face instead — the ghost keeps the base composition (the same $n_{\text{tot},bc} + \delta p_{bc}$ particles at ρ_{bc}) but carries the temperature of the first interior cell,

$$p_{\text{gh}} = (n_{\text{tot},bc} + \delta p_{bc}) T_1, \quad T_1 = \frac{p_1}{(n_{\text{tot}} + n_e)_1}, \quad (8)$$

with $(n_{\text{tot}} + n_e)_1$ (`n_part_cell1`) refreshed by the composition solve, so the ghost pressure stays a differentiable function of the interior pressure — what the JFNK line search needs — while the ionization state it divides by is lagged like every other composition quantity in a hydro step (`Apply_BC.f90`, lines 55–77; `input_read.f90`, lines 588–602). The motivation is that the T_0 pin has no physical backing — the radiative equilibrium of the lower atmosphere is outside the model — and becomes a contact discontinuity sitting on the boundary when the first cell cools well below T_0 . Measured on the production HD 189733 b configuration (`docs/hd189_base_checkerboard.md`, §13): T_{gh} goes from 1183 K to 540 K against $T_1 \approx 494 \rightarrow 540$ K, ρ_1/ρ_{gh} from 2.33 to 0.95, and the alternating $\ln \rho$ amplitude over cells 1–12 from 2.6×10^{-2} to 9.2×10^{-3} , at an unchanged $\log_{10} \dot{M}$ (9.04 vs. 9.05). Note that with Base BC: pressure the microbar target is imposed at T_0 when n_0 is derived, so under a continuous- T ghost only the base *density* remains anchored; `input_read` says so.

Hydrostatic base (Hydrostatic base: True). The legacy ghost pressure is a constant across the two ghost cells, so dp/dr is flattened to ≈ 0 at the base — while steady momentum balance there needs $dp/dr = -\rho g$. This flag replaces the ghost pressure by the linear extrapolation of the *interior* pressure gradient (cells 1 and 2),

$$p_{\text{gh}}(r) = p_1 + \frac{p_2 - p_1}{r_2 - r_1} (r - r_1), \quad (9)$$

so that dp/dr is continuous through the base face and can balance gravity there — the base-face momentum residual being the source of the breathing (§2.4), which for strongly bound planets (large b_0 , e.g. HD 189733 b) is $O(10\text{--}100)$. The density stays anchored at ρ_{bc} ; the base temperature then floats slightly off T_0 , since neither T nor p is fixed any more (`Apply_BC.f90`, lines 55–72).

Low-Mach contact-mode damping (Low-Mach damping: <eps4> [<M_th>], 2026-08-13). Not a boundary condition, but aimed at the same near-base region. The HLLC flux carries the contact/entropy wave at $S_* \sim v$, so the dissipation it applies to that family vanishes as the flow stagnates and a $2\Delta r$ mode in v and T becomes marginally damped — which is what a shell where metal cooling has stopped the flow looks like. The key adds a gated fourth-difference (Jameson–Schmidt–Turkel) stress to the numerical *momentum* flux, together with its work term $v D_p$ in the energy flux so the pair conserves total energy; the mass flux is untouched. The gate closes the term identically above $M > M_{\text{th}}$ (default 10^{-3}). It lives in `src/modules/flux/low_mach_dissipation.f90` and is applied inside `RK_rhs`, the one routine the steady residual assembly also calls, so the marching loop and the Newton residual see the *same* equation — unlike the Shapiro filter, which touches only the marching

state, leaving Newton to solve a system whose mode the filter had been hiding. Off by default ($\varepsilon_4 \leq 0$ skips the code path entirely). *Caveats, both measured* (docs/hd209_metal_stagnation.md, §§8–9): the coefficient has an operating window — `input_read` warns above the explicit stability bound $\varepsilon_4 < 1/(16\text{CFL})$, and on the `examples/16` warm restart only 5×10^{-3} and 10^{-2} gave a clean Newton finish — and on the *cold* start of that same configuration no coefficient helped at all: off, 5×10^{-3} , 10^{-2} and 2×10^{-2} all bottom out at the same $\sim 10^{-3}$ residual floor, so the cold hand-off state appears to lie outside the Newton basin and the damping is not the lever there.

Implementation note. The cell-by-cell ionization-equilibrium sweep is now OpenMP-parallel over cells (the metal and charge-exchange coefficients in each cell are kept thread-private); the parallel result is bit-identical to the serial one, so none of the BC/IC behavior below is affected. See docs/openmp_parallelization.md.

2.7 Temperature, ionization, and metals at the lower boundary

No boundary condition is imposed on the temperature or on any ionization fraction. The ghost cells are included as ordinary cells in the ionization-equilibrium sweep (`ionization_equilibrium.f90` loops over $j = N+N_g, \dots, 1-N_g$), so their T , H/He stages, He 2^3S , and all metal stages come out of the same coupled MINPACK solve as the interior, evaluated at the ghost state $(\rho_{bc}, n_{\text{tot},bc} + \delta p_{bc})$. Combined with §2.2 this makes the effective thermodynamic boundary condition “ $n = n_0$, $T = T_0$, ionization in local equilibrium”. Metal abundances enter only through the runtime `metals.inp` fractions; the boundary needs no special metal treatment: their abundances ride on the (fixed-composition) density BC, and under the default `eos_metals 1` policy their mass, electrons, and nuclei are carried consistently through `calc_rho/calc_ne/calc_ntot` (`eos_metals 0` restores the legacy trace approximation in which they never appear in the mass or momentum budget).

3. Upper boundary: supersonic free outflow

The upper ghost cells are filled by extrapolation (`Apply_BC_W`, lines 95–99):

```
do k = 1, Ng
  W(N+k,:) = W(N,:) ! PLM / first order
  if (use_weno3) W(N+k,:) = 2*W(N+k-1,:) - W(N+k-2,:) ! WENO3
enddo
```

i.e. zeroth-order (copy) extrapolation of all primitive variables for the PLM scheme, upgraded to linear extrapolation when WENO3 reconstruction is active (the wider stencil of WENO3 is sensitive to the artificial flattening a plain copy would imply). `Rec_BC` applies the same rule to the reconstructed interface states.

The justification is characteristic-based: by the time the flow reaches r_{max} the wind is expected to be supersonic, so all three characteristics leave the domain and no physical information may be prescribed from outside; extrapolation is then the consistent “transparent” closure. Two caveats follow directly:

- If the flow is *not* supersonic at $r_{\max}$ (observed for deep Roche-lobe-overflow geometries, e.g. WASP-121b Case D, which is subsonic at L1), one in-going characteristic formally remains and the extrapolation BC is no longer strictly well-posed; in practice the scheme remains stable but the outer solution should be treated with caution (geometry-limited regime).
- The BC carries no information about the medium beyond $r_{\max}$ (stellar wind, confinement pressure); EXHALE models the planetary outflow only.

No ionization or temperature condition is applied at the top either; the ghost cells participate in the equilibrium sweep like all others, with their extrapolated (ρ, p) .

4. Boundary-adjacent settings

These are not boundary conditions of the hydrodynamic update, but they are boundary choices of the model and are easily confused with BCs.

4.1 Radiative transfer: irradiation from the top

The stellar EUV/X-ray flux enters through the *outer* boundary. The column densities that attenuate the ionizing flux are integrated top-down (`calc_column_dens`, `utilities.f90`): the outermost ghost cell is assigned its own local column $N(N+N_g) = n \Delta r$, and the recursion $N(j) = N(j+1) + n_j \Delta r_j$ proceeds inward, i.e. the optical depth is (essentially) zero at the top of the grid and grows toward the base. This is the radiative-transfer analogue of a boundary condition: the unattenuated stellar spectrum is imposed at $r_{\max}$, and the base receives whatever survives the overlying column. Because the base column is huge, the lower boundary is effectively in radiative shadow — consistent with holding it at (n_0, T_0) .

4.2 Escape radius and the convergence window

The `Escape radius [R_p]`: input $(r_{\text{esc}}$, default 1.5) defines $j_{\min}$, the first cell with $r \geq r_{\text{esc}}$ (`define_grid.f90`). All steady-state diagnostics — momentum-flux flatness $\rho v r^2 = \text{const}$, du , dtu — are evaluated over $[j_{\min} : N]$ only, deliberately excluding the breathing base (§2.4). A guard added for deep-RLOF geometries (EXHALE): if $r_{\text{esc}} \geq r_{\max}$ the window would be empty (and `maxval/minval` over an empty section return $\mp \text{HUGE}$, producing a spurious instant “convergence”); in that case the window is clamped to the outer half of the domain and a loud warning asks the user to lower r_{esc} .

2026-08-15: the layer below the window is measured, not accepted silently. The acceptance window is still $[j_{\min} : N]$, but the layer $[1 : j_{\min}-1]$ excluded from it is no longer unreported. `resid_relnorm_below_escape` (`src/modules/time_step/steady_newton.f90`) applies the same volume-weighted relative norm to that layer and locates the cell carrying the largest cell-wise residual in it; `write_resid_below_escape` prints one line at the start and end of the PTC solve and at the start and end of the JFNK solve. The value is printed and never tested — nothing in the solve depends on it.

The momentum row of that diagnostic is *not* scaled by $|u_2| = |\rho v|$. The layer is quasi-hydrostatic ($|v| \sim 10 \text{ cm s}^{-1}$), so that denominator collapses and the ratio is large wherever the flow is slow, whether or not the state is right: converged states read 0.2–10 on it, the metals-off solution nobody doubts reading the largest. The physical scale of the momentum equation there is the hydrostatic pair $|dp/dr| \sim \rho g$, so the momentum row is divided by the gravitational force density

$$s_{\text{grav}}(j) = |\rho_j| \frac{|G\phi_i(j) - G\phi_i(j-1)|}{\Delta r_j}, \quad (10)$$

the same discrete potential difference the momentum source term uses (Source.f90); the printed number is then the fractional violation of hydrostatic balance in the layer. On that scale the converged HD 209458 b solutions — including the metals-off one that read 9.99 before — all hold the layer at 10^{-5} to 1.4×10^{-4} of ρg , comparable to the wind-region Resid tol. The mass and energy rows keep the $|u|$ scale and print separately.

Widening the acceptance window to include the layer was considered and is *rejected by measurement*: on the old $|u|$ scale a converged state carries 0.2–10 there, four to six orders above Resid tol, so the widened window would reject solutions that are correct. Both measurements: docs/hd209_metal_stagnation.md, §§9.4 and 10.

4.3 Domain truncation (where the upper boundary sits)

In Roche mode the grid is truncated at the Hill radius $r_{\text{max}} = (3M_*/M_p)^{-1/3} \tilde{a} \approx L1$; in spherical mode at the user-supplied Outer radius [R_p]:. The choice changes where the free-outflow BC is applied, and hence how defensible the supersonic assumption is (§3.); it does not change the BC itself.

5. Initial conditions (set_IC.f90)

Three concepts to fix first: cold IC, warm IC, and starter. For the reader's convenience, every IC family below is a variation on three ideas. The distinction matters because it controls whether the run starts *inside* the wind basin or has to be pushed there.

Cold IC. The default. A fully *neutral*, isothermal ($T = T_0$) *hydrostatic* atmosphere with only a small outward velocity seed. Nothing is pre-heated or pre-ionized: the simulation grows the temperature, ionization, and wind from scratch. Because it is built from EXHALE's own base BC (fixed ρ_{bc}, T_0), it is automatically *consistent with the lower boundary* and relaxes cleanly to the transonic wind.

Warm IC (warm seed / warm start). An initial state that already carries a *heated, ionized, flowing* profile, overlaid on the hydrostatic density — the transonic isothermal-wind and hot-Parker families. A plausible wind therefore exists from step 0, which is needed where a cold start cannot launch (deep-RLOF or low-gravity boil-off). It is still EXHALE's own base with a thermodynamic overlay, so it stays consistent with the lower boundary; it only auto-enables force_start so the heating can settle.

Starter (starter solution / seed). A *complete, externally pre-converged* steady wind solution read in wholesale as the IC — the Wind-AE warm start (IC mode: windae), which ramps a Wind-AE windsoln (or a grid/database seed) to this planet. Unlike a warm seed, a starter *replaces the entire state* with another code's solution, including its

own base; that base can clash with EXHALE’s lower BC and pull the run off the wind branch (§7.). A starter bootstraps hard cases but is not guaranteed consistent with EXHALE’s boundary.

When a run does not restart from saved profiles (§5.5), the initial state is built by `set_IC` in `src/modules/init/`. Five options are available: the three IC families `set_IC` actually constructs (cold hydrostatic, transonic isothermal wind, hot-Parker warm seed), plus `IC mode: auto`, which selects among them at run time, and `IC mode: windae`, whose profile is built in `init.f90` from the Wind-AE solver. The choice is made by optional `input.inp` keys, with the cold hydrostatic atmosphere as the default:

Family	Key	Default
Cold hydrostatic + velocity seed	(none)	yes
Transonic isothermal wind	Transonic IC: True	off
Hot-Parker warm seed	Hot Parker IC: T_{wind} [K]	off
Automatic selection	IC mode: auto	off
Wind-AE warm start	IC mode: windae	off

The transonic and hot-Parker families — and only those two — force the first 1000 iterations (`force_start`) so that the heating structure can develop before the convergence test is allowed to fire. Whatever the family, the full dimensional IC is dumped to `output/IC_dump.txt` for inspection before the first step.

Automatic selection (IC mode: auto). `select_IC_auto` (`set_IC.f90`) picks the family from a single exact probe — `find_sonic` at the *cold* base sound speed in the actual potential. An interior sonic point selects the transonic IC; this catches both deep-RLOF bases (the Roche $\phi' \rightarrow 0$ topology toward L1 always crosses the critical condition; e.g. WASP-121b Case D, $r_c = 1.30$) and low-gravity boil-off planets ($r_c \simeq b_0/2c_0^2$ falls inside the domain for small b_0). Otherwise the cold hydrostatic default is kept (classic EUV-heated winds: HD 209458 b has $r_c \sim 54 R_p$, far outside the domain). There is no tunable threshold; b_0 (= the escape parameter Λ) is logged as a regime diagnostic, the warm seed is never auto-selected (the benchmark showed it buys nothing when the cold start works), and the explicit keys override auto. Design rationale and validation: `docs/auto_ic_design.md`.

Wind-AE warm start (IC mode: windae). With this key, `init.f90` builds the initial condition by running the in-tree Fortran port of the Murray-Clay/Broome *Wind-AE* steady-state wind solver (`src/modules/wind_ae/`). The bridge `wae_exhale_bridge` maps the planet’s EXHALE parameters ($M_p, R_0, M_*, a, F_{\text{XUV}} = J_{\text{XUV}}, L_*, \text{He/H, domain mode}$) to Wind-AE, ramps a shipped seed solution (`inputdata/windae_seed.csv`) to the planet by continuation, runs the BVP relaxation + outward integration, and writes `output/{Hydro_ioniz, Ion_species}_IC.txt` on the EXHALE grid; `load_IC` then ingests them. The whole EXHALE-input $\rightarrow$ Wind-AE solve $\rightarrow$ IC $\rightarrow$ run is a single EXHALE.x invocation. This is a metal-free H/He *warm start* (EXHALE re-equilibrates the ionization and any metals from the first step); finish with `Solver: Newton` for a quantitative $\dot{M}$. Best suited to seed-adjacent hot Jupiters; far-from-seed planets may fail the ramp (the run then advises `IC mode: auto`). A standalone generator `wind_ae_ic.x` (`make wind_ae_ic`) does the same out of process. Full port/validation: `md_backup/wind_ae_fortran_port_plan.md`.

The guiding principle in all three cases is *consistency with the lower boundary condition* of §2.: every IC matches $(\rho, p, v) = (\rho_{\text{bc}}, n_{\text{tot, bc}} + \delta p_{\text{bc}}, 0)$ at the base exactly, so the boundary never has to fight the initial state. The quantitative

comparison of the families is in docs/initial_condition_benchmark.pdf; the conclusions quoted below come from there.

5.1 Default: cold neutral hydrostatic atmosphere + velocity seed

The original ATES initial condition, kept as the EXHALE default because it is the most robust (§5.3).

Density. An isothermal hydrostatic atmosphere in the (Roche or spherical) potential, anchored to the base density BC:

$$\rho(r) = \rho_{bc} \exp \left[b_{0,\text{eff}} (\phi(r_0) - \phi(r)) \right], \quad (11)$$

with one deliberate distortion: $b_{0,\text{eff}}$ starts at the physical value (1 in code units, the potential already carrying the Jeans parameter b_0) and is increased in steps of 0.2 until the density at the domain midpoint $r_{1/2} = (1 + r_{\text{max}})/2$ falls below 10^{-8} (set_IC.f90, lines 85–101). For low-gravity planets the true isothermal atmosphere would not decay enough across the domain; the loop artificially *steepens* the initial atmosphere so that the outer half of the grid starts effectively empty. The profile is then floored at $\rho = 10^{-8}$ (line 112). Equation (11) with $b_{0,\text{eff}} > 1$ is therefore *not* an equilibrium — it is a concentrated cold reservoir that the subsequent heating will evaporate.

Velocity. A small outward linear seed,

$$v(r) = \frac{1}{2} (r - r_0), \quad (12)$$

zero at the base and growing outward (line 108). Starting from exact rest delays the launch of the wind; the seed gives the outflow a direction without imposing any particular magnitude.

Pressure, temperature, ionization. Isothermal at the base temperature and consistent with the lower BC (lines 137–146):

$$p = (n_{\text{tot},bc} + \delta p_{bc}) \frac{\rho}{\rho_{bc}}, \quad T = 1 (= T_0 \text{ everywhere}), \quad (13)$$

and a mostly neutral composition: $x_{\text{HII}} = \delta p_{bc} \sim 10^{-10}$, $x_{\text{HeII}} = 10^{-10} Y$, HeIII and He 2³S zero, the rest neutral. At the base cell this reproduces $(\rho_{bc}, n_{\text{tot},bc} + \delta p_{bc}, T_0)$ exactly.

The physical picture: *a cold, neutral, hydrostatically stratified atmosphere is exposed to the stellar EUV flux at $t = 0$ and left to evaporate*. The first ionization-equilibrium sweep deposits the heating, the upper layers warm and expand, and the wind launches self-consistently — typically within the forced-start phase.

5.2 Transonic isothermal-wind IC (EXHALE)

Motivation: deep Roche-lobe-overflow geometries (the canonical case is WASP-121b Case D, whose sonic point sits essentially at L1) launch *no* wind from the hydrostatic IC — the run stalls in a bound, breathing state. For these cases the run must start from a flowing solution. With Transonic IC: True, wind_profile (set_IC.f90, lines 177–233) solves the steady isothermal wind in the actual ATES potential via its Bernoulli integral:

$$\frac{1}{2} v^2 - c^2 \ln v - 2c^2 \ln r + \phi(r) = B, \quad c^2 = \frac{n_{\text{tot},bc} + \delta p_{bc}}{\rho_{bc}}, \quad (14)$$

where c is the dimensionless isothermal sound speed of the cold base gas. The constant B is fixed at the sonic point r_c , located by bisection on the critical condition

$$\phi'(r_c) = \frac{2c^2}{r_c} \quad (15)$$

(find_sonic; in Roche mode $\phi' \rightarrow 0$ toward L1, so a sign change is generic). At each radius Eq. (14) has one subsonic and one supersonic root; the transonic wind takes the subsonic branch for $r < r_c$ and the supersonic branch for $r > r_c$ (robust bisection per cell, wind_root). The density then follows from steady mass conservation pinned to the base BC,

$$\rho v r^2 = \rho_{bc} v(r_0) r_0^2. \quad (16)$$

If no interior sonic point exists (gravity dominates out to $r_{\max}$: a bound “breeze” case), the routine reports failure and set_IC falls back to the hydrostatic IC of §5.1 with a console notice. If the base itself is already super-critical, the flow is taken supersonic from the base.

5.3 Hot-Parker warm-seed IC (EXHALE)

This family is the product of the initial-condition benchmark (docs/initial_condition_benchmark.pdf, §7), whose central negative result deserves restating: **seeding the full isothermal Parker wind — its density and velocity — makes the run diverge**. The hot, low-density Parker column is globally inconsistent with the non-isothermal ATES equilibrium; it drains onto the base and drives the self-amplifying breathing mode of §2.4 (whereas the cold hydrostatic IC breathes self-limitingly and converges). The lesson: the base may not be handed an initial state whose global column it cannot support.

The robust compromise, enabled by Hot Parker IC: T_{wind} (default $T_{\text{wind}} = 10^4$ K), is a *warm seed*: keep the stable cold hydrostatic *density* of §5.1 and overlay only the fields whose relaxation is slow and stiff — temperature, ionization, and a velocity head-start. The overlay is anchored to the cold base by a spatial smoothstep window (lines 114–133),

$$s(r) = \text{clip}\left(\frac{r-1}{r_{\text{tr}}-1}, 0, 1\right), \quad \xi(r) = s^2(3-2s), \quad (17)$$

with $r_{\text{tr}} = \text{hp_base_rtr} = 1.3 R_p$ (a fixed default in parameters.f90; it is a module variable, not a Fortran parameter, but no input key exposes it), so that $\xi = 0$ at the base and $\xi = 1$ for $r \geq r_{\text{tr}}$. The overlaid fields are

$$\begin{aligned} T(r) &= 1 + \left(\frac{T_{\text{wind}}}{T_0} - 1\right) \xi, & v(r) &= v_p(r) s, \\ p(r) &= \rho \frac{n_{\text{tot},bc} + \delta p_{bc}}{\rho_{bc}} T(1 + \xi), & x_{\text{HII}}, x_{\text{HeII}} &: \text{neutral} \rightarrow \text{ionized as } \xi, \end{aligned} \quad (18)$$

where the $(1 + \xi)$ factor in the pressure is the electron partial pressure of the ionized warm gas (particle number doubles at full ionization), and v_p is the Parker velocity head-start: the transonic profile of §5.2 re-solved with the *warm, ionized* sound speed

$$c_{\text{warm}}^2 = \frac{n_{\text{tot},bc} + \delta p_{bc}}{\rho_{bc}} \frac{T_{\text{wind}}}{T_0} \times 2, \quad (19)$$

its *density discarded* (that is the part that destabilizes), and its velocity multiplied by s so it vanishes at the base. If the warm profile has no interior sonic point, the head-start quietly degrades to the small linear seed.

At the base $\xi = 0$ gives $T = T_0$, $p = n_{\text{tot},bc} + \delta p_{bc}$, $v = 0$ — the lower BC of §2. exactly, with no inverted gradients for the first step to undo.

Benchmark verdict. The warm seed converges to the same $\dot{M}$ as the cold IC with no NaN/blow-up, but yields *no speedup* (−2.3% steps in the benchmark): ATES convergence is gated by the slow momentum (du) relaxation, not by the thermal/ionization relaxation the warm seed shortcuts. The cold hydrostatic IC therefore remains the default; the warm seed is kept for cases where the cold start struggles to launch.

5.4 Metals and common finalization

In every family the metals start maximally neutral: the entire elemental abundance is placed in the neutral ground stage,

$$f_{X^0} = \frac{n_X/n_H}{1+4Y}, \quad f_{X^+} = f_{X^{++}} = 0 \quad (20)$$

(set_IC.f90, lines 152–162; abundances from metals.inp). The first ionization-equilibrium sweep redistributes each element over its stages immediately, so the choice costs at most one equilibrium solve. The complete IC (dimensional r, n, v, p, T) is written to output/IC_dump.txt before the first step.

5.5 Restarts (Load IC)

When Load IC is enabled, set_IC is bypassed and the previous run’s profiles (including ghost cells) are read back from the *_IC.txt copies (load_IC.f90). This only sets the initial state; Apply_BC overwrites the ghost cells from the restored interior at the first step, so a restart cannot smuggle in a different boundary condition. (Operationally: the output/ → *_IC.txt copy is performed by the GUI, not by the Fortran code — for scripted runs the *_IC.txt files must be seeded by hand, or the code stops on the empty files it creates.)

2026-08-13: a restart never carries zero metals into a metals-on base. The schema-2 writer emits the metal columns unconditionally, so testing only for the *presence* of a column accepted a file written by a metals-off run and let the restart proceed with zero metal density everywhere — while the lower boundary still counts the metals in the mass and particle budget under eos_metals, leaving an $O(1)$ residual in the first cells. load_IC now treats an element as absent in either case: no column for some stage, *or* columns that sum to zero over the interior. Such an element is rebuilt exactly as a cold start builds it, $n_X = a_X n_H$ with all of it neutral, and the substitution is reported; the rebuild happens *before* calc_rho reconstructs the mass density, so the restart keeps the same mass closure the cold start has by construction. The elements that were substituted are recorded in memem_from_abundance and echoed by the setup report.

2026-08-15: a restart requires an IC written at the same N . Since the cell count is a runtime key (§1.), an IC file written at a different N would previously have died in the middle of the read with a bare end-of-file error. load_IC now counts the data records of output/Hydro_ioniz_IC.txt first — the files carry one record per cell, $N+2N_g$ rows — and stops with an explicit message naming both counts when they disagree, so a mismatched Grid cells: and a truncated file are reported rather than guessed at. There is no interpolation onto a new grid: a restart requires an IC written at the same N .

2026-08-27: a restart lands on the reservoir the handoff states. A restart file carries the metal densities of the state it was written from. When the reservoir itself has moved since — which is what a flux-closure iteration does to the elemental ratios of the lower atmosphere — the base boundary condition would use the new El/H while the loaded column above it still held the old one, and the elemental budget $n_{\text{El}}/n_{\text{H}} = (\text{El}/\text{H})_{\text{resolved}}$ could not close above the first cell. `load_IC` therefore rescales the whole column of an element *the handoff itself states* by the single factor

$$r_{\text{El}} = \frac{(\text{El}/\text{H})_{\text{handoff}}}{(\text{El}/\text{H})_{\text{loaded, base cell}}}, \quad (21)$$

the same factor for every ionization stage, so the loaded ionization split and the radial shape survive and only the normalization changes; the factor and both ratios are printed. “States” means the one door `set_element_abundance` guards, which the `<El>_H_base` keys of `base.inp` and the elemental ratios of a lower-atmosphere profile both go through, so an abundance that came from `metals.inp` alone — and every restart with no handoff at all — leaves the loop doing nothing and is bit-for-bit the restart the earlier code produced. El/H at the base cell is counted in nuclei, the element summed over its stages against the hydrogen nuclei (free plus the H bound in H_2 , H_2^+ , H_3^+ and HeH^+). Helium keeps its own convention — the base cells are set to the input He/H while the column above may carry a diffused split — and is not rescaled here.

6. Summary

Quantity	Lower boundary (base, $r = 1$)	Upper boundary ($r = r_{\text{max}}$)
Density ρ	Dirichlet: $\rho_{\text{bc}} = (m_{\text{gas}}/\text{H})/(1+Y)$ (nuclei = n_0)	copy / linear extrapolation
Velocity v	one-way valve $\max(v_1, 0)$ (smooth / mass-flux opt.)	copy / linear extrapolation
Pressure p	$n_{\text{tot, bc}} + \delta p_{\text{bc}}$, $\delta p_{\text{bc}} = n_e(\text{base})/n_0$	copy / linear extrapolation
Temperature T	implied: $T = T_0$ exactly (§2.2)	implied by extrapolated (ρ, p)
Ionization, metals	equilibrium solve in ghosts (no BC)	equilibrium solve in ghosts (no BC)
Radiation	in radiative shadow (large column)	unattenuated stellar spectrum
Type	isothermal–isochoric reservoir + valve	supersonic free outflow
Code	<code>Apply_BC.f90:31--80</code>	<code>Apply_BC.f90:95--99</code>

Table 1: Boundary conditions at a glance, with every option off (the defaults). Ghost cells: $N_g = 2$ per side; BCs act on ghosts only and reach the interior through the HLLC fluxes. $m_{\text{gas}}/\text{H} = 1 + 4Y + \sum_X a_X A_X$ is the gas mass per H nucleus from composition. `f90`, Eq. (3); it reduces to the legacy $1 + 4Y$ for a metals-off atomic base. The velocity and pressure rows have the opt-in alternatives of §2.6: mass-flux base velocity, and a ghost pressure set by dp/dr continuity (Hydrostatic base) or by $dT/dr = 0$ (Base ghost temperature: continuous), the latter two replacing the implied $T = T_0$.

The lower boundary is best read as a single physical statement — *the wind is rooted in an infinite isothermal reservoir at (n_0, T_0) that supplies but never absorbs mass* — implemented as fixed ρ , electron-corrected fixed p , and a one-way velocity valve. Its characteristic dynamical signature is the self-limiting breathing oscillation of §2.4, which is excluded from convergence diagnostics via r_{esc} , smoothed for the Newton solver via `Valve_eps`, and guarded against

in the advection post-processor. The upper boundary is a plain supersonic free-outflow extrapolation whose validity rests on the wind crossing the sonic point inside the domain.

The initial conditions are built to respect the same reservoir: all three families (§5.) match the base state $(\rho_{bc}, n_{tot,bc} + \delta p_{bc}, v=0)$ exactly, and the benchmark experience condenses into one rule — *the density column must be one the base can support*. The cold hydrostatic atmosphere (default) satisfies this trivially and evaporates into the wind on its own; the transonic IC replaces it with a flowing solution where the hydrostatic start cannot launch (deep-RLOF, sonic point at L1); the hot-Parker warm seed may overlay temperature, ionization, and velocity on the cold density, but seeding the Parker *density* itself tips the breathing base from self-limiting to self-amplifying and crashes the run.

7. Empirical base-BC test matrix (HD209458b)

7.1 Convergence criterion: history and the flux-based decision

The convergence test used here changed over the course of this work, and it is worth stating plainly where it landed and why.

- **Before the Newton finish — flux criterion.** The original time-marching run (PLM, then two-stage PLM→WENO3) was stopped on the *flux* criterion: the fractional variation of the mass flux, $\Delta\dot{M}/\dot{M} < 10^{-3}$, where $\dot{M} \propto \rho v r^2$ (the code’s du is precisely the radial spread of $\rho v r^2$ over the wind). This is the ATES criterion of Caldiroli et al. (2021) and is equivalent to the CETIMB requirement (Koskinen et al. 2013a) that the flux constant $F_c = \rho v r^2$ be constant with altitude.
- **With the Newton finish — a stronger residual criterion.** When the JFNK steady solver was added, it was driven to a much stricter target: the steady residual $\|R\| = \|du/dt\| < 10^{-3}$ on *every* conserved equation (mass, momentum, energy). This residual norm is far more demanding than flux flatness, and at the time these runs were made the JFNK solver reached it on only some of them; several full-physics runs were labeled “JFNK fail” even though their wind was perfectly flux-converged. *(2026-08-10: the “JFNK fail” outcomes recorded in this section were produced with a diagonal scaling whose momentum floor was set by the base cell, and with a watchdog that aborted after 15 iterations without a new best $\|R\|$ — before the opening pseudo-transient has finished raising it. Both have since been replaced and the same configurations converge; see docs/newton_scaling_and_base_wall.md. The residual-vs-flux decision below is unaffected, but the “held up by the near-base momentum” attribution attached to individual rows is withdrawn.)*
- **The reference codes use only the flux criterion.** Neither ATES (Caldirola 2021: $\Delta\dot{M}/\dot{M} < 10^{-3}$) nor CETIMB (Koskinen 2013a: F_c constant with altitude) imposes a residual-norm threshold. The reference standard for “the wind has converged” is flux flatness; the residual norm is an EXHALE addition tied to the optional Newton finish.
- **Decision (adopted).** The **convergence is judged by the flux criterion**, matching the reference codes; the steady residual $\|R\|$ is computed and reported *for reference only* and gates the stop solely when the user explicitly

requests it (`Resid_tol:`). In the code the default monitor is flux-based ($du < du_{th}$ plus mass-flux level stability) and the residual — volume-weighted by default (`resid_vol` in `parameters.f90`), since the $L-\infty$ max is inflated by the small near-base cells — is printed each `N_resid` steps as a diagnostic. The verdict column of the table below therefore reads `flux-conv`, not “JFNK fail”, for runs that are flux-converged but whose residual finish did not reach the target.

- **2026-08-11: the du triggers are armed on an ascending crossing.** du is the radial spread of $\rho v r^2$, so a *small* du is evidence of relaxation only for a state the marching loop has actually relaxed. A freshly generated IC — cold hydrostatic, transonic, Wind-AE, hot Parker — is an analytic profile whose du measures the smoothness of the formula rather than the flatness of a wind: the WASP-121 b transonic IC measured $du(1) = 1.85 \times 10^{-5}$, which tripped both the Newton hand-off and the secondary-ionization flip on step 1 and ran away to NaN. Each du -keyed trigger — the $du < du_{th}$ stop, and the JFNK hand-off that the secondary-ionization flip shares — is therefore *armed* only once du has been seen at or above its own threshold, and fires only on the way back down (`du_stop_armed`, `du_newton_armed` in `EXHALE_main.f90`). A state read back with `Load IC? True` was relaxed by the marching loop that wrote it, so its du is meaningful and both triggers start armed. Cold hydrostatic starts arm on step 1 anyway (measured $du(1) = 0.37$ for the WASP regression cases), so their behavior is unchanged. Details: `docs/newton_scaling_and_base_wall.md`, §9.
- **2026-08-15: the Wind-AE IC measured under that guard.** The arming guard had been measured on the transonic and cold-hydrostatic ICs but not on IC mode: `windae`, whose ~ 3 -step false du stop is what originally forced the “always Newton-finish” rule. Measured on `examples/12_windae_ic_hd209` with a 2000-step cap: $du(\text{step } 2) = 1.43 \times 10^{-4}$, below every threshold on the untouched generated IC, and nothing fires; the du stop arms at step 23 ($du = 1.027 \times 10^{-3}$) and the Newton hand-off at step 222 ($du = 1.002 \times 10^{-2}$), both on the first ascending crossing. The false stop is structurally gone for this IC mode too. This does *not* retire the rule of thumb: a du -stop $\dot{M}$ still carries the usual path-dependent spread, so the Newton finish stays the prescription for a quantitative rate.

2026-08-11: the tabulated rates are a product of the solver and the cooling as they stood when the matrix was run. Two changes since then move the numbers in the two tables below, and neither has been re-measured on this matrix. (i) The JFNK line search now accepts, stores, and judges convergence on the true residual — with the WENO smoothness weights recomputed at each trial state rather than frozen at the previous iterate — so runs that were recorded here as failing their residual finish are not evidence about the current solver. (ii) The two ground-term fine-structure coolants of the time — the set has since grown to the eight C I/C II/N II/O I lines — are allowed to be optically thick, and the CHIANTI coronal fits are switched off smoothly below their 10^3 K validity floor; on a planet whose base falls out of the coronal regime this changes the base equilibrium temperature by hundreds of kelvin. The tables are kept as the dated record of the base-BC comparison they were made for, and the *relative* ordering across base modes is what they are being cited for. They should not be read as current mass-loss rates: the production HD 209458 b run under the present code gives $\log_{10} \dot{M} = 9.47$ at $\|R\| = 1.8 \times 10^{-4}$ (the 2026-08-19 production set; it read 9.31 at 7.7×10^{-4} when this paragraph was written on 2026-08-11). These runs were not repeated, because repeating them is a re-measurement

of the whole matrix rather than a correction to it.

To pin down how the lower-boundary choices above actually affect a converged wind, HD209458b was relaxed under a controlled combination of solver and base options (the runs live in EXHALE/HD209458b_test/). Each folder name encodes the model as <base>_<solver>_He{0,1}_met{0,CNO,FULL}[_flags]: **base** is dens ($n_0 = 10^{14} \text{ cm}^{-3}$ fixed; the dens12p5 run lowers it to $\log n_0 = 12.5$) or pres1ub (pressure-anchored at $1 \mu\text{bar}$, §2.6); **solver** is PLM (single-stage, stopped at a du plateau), PWN (two-stage PLM→WENO3 then a Newton/JFNK finish), or PLMN (single-stage PLM then JFNK, skipping WENO3); **S/V** are the Shapiro filter / mass-flux base velocity of §2.6; **He/met** are He 2^3S and the trace metals (C,N,O or the full C,N,O,Mg,Ca,Na,Fe set); **IC** is the cold hydrostatic or the Wind-AE warm start (§5.). The results are split into a pressure-anchored-base table (Table 2) and a density-anchored-base table (Table 3).

In both tables: $\dot{M}$ is in g s^{-1} , v_{out} in km s^{-1} , and neg.% is the *radius-weighted* infall fraction — the fraction of the linear radius range $[r_1, r_N]$ over which $v < 0$ (each cell weighted by its width Δr). This is used in place of the raw cell-count fraction, which the non-uniform grid (cells packed near the base) inflates: the dense near-base sliver of $\text{sub-}10 \text{ ms}^{-1}$ noise would otherwise read as tens of percent. The radius-weighted measure cleanly separates that base noise ($\lesssim 4\%$) from genuine infall (tens of percent). “—” marks a run on an infall branch (wind $\dot{M}$ undefined).

The **verdict** column reports convergence *by the flux criterion* (the $\Delta\dot{M}\%$ column), *not* by the residual: Newton = a genuine Newton-converged steady state (solver info=0, $\|R\| < 10^{-3}$); flux-conv = flux-converged ($\Delta\dot{M}\% < 1$) but the JFNK residual finish did not reach $\|R\| < 10^{-3}$ — by the ATES/CETIMB standard these *are* converged winds, not failures; marginal = $\Delta\dot{M}\% \sim 1\text{--}5\%$ (a wind, not tightly flat; carries the $\sim 2\text{--}5\%$ path-dependent $\dot{M}$ spread of the PLM-stall runs); not conv = $\Delta\dot{M}\% > 5\%$ (an unconverged / breathing wind); infall = relaxed to an accretion branch. $\Delta\dot{M}\%$ is the radial spread of the mass flux $\rho v r^2$ over the wind region ($r \geq r_{\text{esc}}$), $(\text{max} - \text{min}) / \text{mean}$ — this is the actual **convergence criterion of the reference codes**: ATES (Caldioli 2021) stops at $\Delta\dot{M} / \dot{M} < 10^{-3}$, and CETIMB (Koskinen 2013a) requires $\rho v r^2 = F_c$ to be constant with altitude. We mark $\lesssim 1\%$ as *flux-converged* (a genuine steady wind *by the reference standard*, independent of the $\|R\|$ /Newton outcome); “—” = infall (not a wind). Note that all PWN rows are flux-converged ($\leq 0.9\%$) even where the stricter $\|R\|$ test labels them “JFNK fail”. The “JFNK fail” labels date from the pre-2026-08-10 solver; the flux verdicts do not depend on them.

folder	BC	solver	S	V	He	met	IC	$\log \dot{M}$	v_{out}	neg.%	$\Delta \dot{M}\%$	verdict
pres1ub_PWN_He0_met0_cold	pres	PWN	–	–	–	–	cold	10.39	+8.59	0.0	0.53	Newton
pres1ub_PWN_He1_metCNO_cold	pres	PWN	–	–	✓	CNO	cold	9.53	+6.63	1.6	0.62	flux-conv
pres1ub_PWN_He1_metFULL_cold	pres	PWN	–	–	✓	FULL	cold	9.37	+6.29	2.2	0.87	flux-conv
pres1ub_PWN_He1_metFULL_cold_duJFNK	pres	PWN ^(d)	–	–	✓	FULL	cold	9.37	+6.29	1.7	0.88	flux-conv ^(d)
pres1ub_PLMN_He1_metFULL_cold	pres	PLMN	–	–	✓	FULL	cold	9.40	+6.10	1.8	3.5	marginal [§]
pres1ub_PWN_He1_metCNO_cold_Vflux	pres	PWN	–	✓	✓	CNO	cold	9.53	+6.63	1.6	0.74	flux-conv
pres1ub_PWN_He1_metCNO_cold_Shap	pres	PWN	✓	–	✓	CNO	cold	10.21	+7.45	25.3	114	no conv. [†]
pres1ub_PLM_He0_met0_cold	pres	PLM	–	–	–	–	cold	10.40	+8.36	0.0	3.5	marginal

Table 2: **Pressure-anchored base** ($1 \mu\text{bar}$, $n_0 \approx 5 \times 10^{12} \text{cm}^{-3}$). Columns and general notation as described in the text. [†]The pressure + Shapiro run never converged: it ran to the 10^6 -step cap with 25% of the radius in infall (vs. 1.6% for the same run without Shapiro), so the filter both prevents convergence and deepens the infall; the quoted $\dot{M}$ is the outer-region value, not a steady state. [§]PLMN = single-stage PLM $\rightarrow$ JFNK (the WENO3 stage skipped). It reaches an *essentially identical final state* to the PWN (PLM $\rightarrow$ WENO3 $\rightarrow$ JFNK) row above — $\log \dot{M} = 9.40$ vs. 9.37 , $v_{\text{out}} = +6.10$ vs. $+6.29$, overlapping $T/v/n/p$ profiles — in about *half* the steps (9.4×10^4 vs. 1.7×10^5); both still fail JFNK and stall, so the WENO3 stage is not the JFNK bottleneck. ^(d)The _duJFNK row is the same full-metal case but with the *flux*-keyed JFNK hand-off (Solver: Newton, JFNK fired once the wind reached $du < 10^{-2}$ rather than $\|R\| < 0.05$). With this hand-off *plus* the volume-weighted residual norm the JFNK drove $\|R\|$ down to 2.3×10^{-3} — a $\sim 100\times$ improvement over the 0.17 it reached with the $\|R\|$ -keyed hand-off — but it stalled there ($\|R\|$ would not drop below $\approx 2 \times 10^{-3}$ for 15 iterations, the holdout being the base cell $j=1$ momentum), so it still falls back and the *final wind is identical* to the metFULL PWN row above ($\dot{M}$, v_{out} , $\Delta \dot{M}\%$ all unchanged). It confirms that the only barrier to a genuine $du < 10^{-3}$ / $\|R\| < 10^{-3}$ convergence for full physics is the localized near-base momentum (§2.4), not the solver path.

folder	BC	solver	S	V	He	met	IC	$\log \dot{M}$	v_{out}	neg.%	$\Delta \dot{M}\%$	verdict
dens_PWN_He0_met0_cold	dens	PWN	–	–	–	–	cold	10.50	+8.80	0.0	0.41	Newton
dens12p5_PWN_He0_met0_cold	dens	PWN	–	–	–	–	cold	10.38	+8.58	0.0	0.42	Newton
dens_PWN_He1_metCNO_cold	dens	PWN	–	–	✓	CNO	cold	9.61	+6.79	2.3	0.54	flux-conv
dens_PWN_He0_met0_windae_spherical	dens	PWN	–	–	–	–	windae	10.37	+5.77	1.4	55	not conv [‡]
dens_PLM_He0_metCNO_cold	dens	PLM	–	–	–	CNO	cold	9.72	+6.77	3.9	4.2	marginal
dens_PLM_He0_met0_cold_Vflux	dens	PLM	–	✓	–	–	cold	10.50	+8.58	0.0	3.5	marginal
dens_PLM_He0_met0_cold_Shap	dens	PLM	✓	–	–	–	cold	—	−13.34	100	—	infall
dens_PLM_He0_met0_cold_Shap_Vflux	dens	PLM	✓	✓	–	–	cold	—	−13.32	100	—	infall
dens_PLM_He0_met0_windae_spherical	dens	PLM	–	–	–	–	windae	10.08	+3.68	0.6	5.1	not conv
dens_PLM_He0_met0_windae_spherical_Shap_Vflux_Force	dens	PLM	✓	✓	–	–	windae+F	—	−20.03	95	—	infall

Table 3: **Density-anchored base** ($n_0 = 10^{14} \text{ cm}^{-3}$; dens12p5: $\log n_0 = 12.5$). Columns and general notation as described in the text. [‡]Wind-AE warm start: JFNK never triggered ($\|R\|$ stayed above the switch); the Wind-AE IC marched under WENO3 drifts to an unconverged breathing state ($du \sim 0.7$), mostly outflow — it does *not* give a clean Newton solution.

What the matrix shows.

- **The Shapiro filter drives infall.** On the cold-IC density runs, every S-on case relaxes to a global infall ($v < 0$ over $\approx 100\%$ of the radius) while the matched S-off case is a clean outflow, independent of the velocity BC — the headline reason the filter is opt-in (§2.6). On the pressure + full-physics case the filter instead prevents convergence altogether (row †).
- **The base BC barely changes the converged wind — the lever is the base density.** With a Newton finish, density ($n_0 = 10^{14}$) and pressure ($1 \mu\text{bar}$) bases give nearly the same $\dot{M}$ (10.50 vs. 10.39). A *density*-anchored run at the same low density (dens12p5, $\log n_0 = 12.5$) reproduces the pressure result almost exactly (10.38 vs. 10.39; +8.58 vs. +8.59), confirming that pressure-anchoring acts only through the lower base density, not through the anchoring variable itself. The mass-flux velocity (V) is benign (it reproduces the valve result).
- **Physics lowers $\dot{M}$.** He 2^3S + C/N/O cooling drops $\log \dot{M}$ from ~ 10.4 to ~ 9.5 – 9.6 ; the full metal set (adding Mg/Ca/Na/Fe line cooling) lowers it further to 9.37. With full physics the JFNK solve fails for both bases and falls back to a stall.
- **Skipping WENO3 before the Newton finish (PLMN) changes nothing but the cost.** On the full-metal case, PLMN (PLM→JFNK) reaches an essentially identical final state to PWN (PLM→WENO3→JFNK) — $\log \dot{M} = 9.40$ vs. 9.37, overlapping profiles — in about half the steps, and both still fail JFNK and stall. So the reconstruction path is not the JFNK bottleneck; the stiff coupled (He 2^3S + 7-metal) residual is. A genuine Newton finish for full physics would need a smoother residual (tighter chemistry tolerance) or a relaxed target, not a different reconstruction.
- **The Wind-AE warm start does not help here.** It either du-stops at step 2 (returning $\approx$ the IC) or, with a Newton finish, never triggers JFNK and drifts to an unconverged breathing state. The cold hydrostatic IC, consistent with EXHALE’s own base, is the reliable choice.

7.2 Assessment: does the Wind-AE starter help?

So far, for HD209458b, the Wind-AE starter does not help — the cold IC is better. Across every Wind-AE-seeded run in Table 3 the outcome is one of: a *du*-stop at step 2 that simply returns the seed (no real relaxation), an unconverged breathing state when a Newton finish is requested (JFNK never triggers), or, when marching is forced, a drift to infall ($v_{\text{out}} = -20 \text{ km s}^{-1}$). An earlier scratch run converged instead to a global *infall root*. None reached a clean, Newton-converged outflow, whereas the cold IC with a PWN finish does (solver info=0, clean wind).

Why. A *starter* replaces the entire state, *including its own base*, with another code’s solution (§5.). That imported base is generally inconsistent with EXHALE’s fixed lower boundary (ρ_{bc}, T_0); the mismatch perturbs the near-base flow off the transonic-wind saddle, and since that saddle is unstable, the run is pulled to the nearby attractors — infall, or the breathing limit cycle. This is the same dense-base/lower-BC tension that drives the breathing (§2.4), now triggered by the seed rather than by a bad global IC.

In fairness. This does *not* make Wind-AE useless. It was brought in for the *hard* cases where a cold start fundamentally cannot launch a wind — deep Roche-lobe-overflow geometries (e.g. WASP-121b Case D, with the sonic point essentially at L1) and low-gravity boil-off planets — where it is, at present, the only way to obtain a transonic seed. HD209458b is an easy case for which the cold IC already works, so there the starter is both unnecessary *and* mildly harmful through the base mismatch. The honest statement is therefore: *the starter is unnecessary for easy cases and, for its intended hard cases, has not yet delivered a clean solution because the base-BC reconciliation is unsolved* — not that the approach is without value.

What would make it useful. The base mismatch must be removed. Two plausible routes (untested): (i) have the Wind-AE bridge hand off only a *warm overlay* — temperature, ionization, and velocity — on top of EXHALE’s own hydrostatic density and base, i.e. demote the “starter” to a “warm seed” (§5.) that stays consistent with the lower boundary; or (ii) ramp the imported base onto EXHALE’s (ρ_{bc}, T_0) boundary before releasing the hydrodynamics. Either could keep the transonic-seed advantage Wind-AE offers for the RLOF cases while avoiding the infall/breathing seen here.

References

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